

MJ2500 – Large Scale Solar Power

Design of a large-scale solar power plant in Brazil

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SolBrilhar

Shine Brighter. Together.

EXECUTIVE SUMMARY

A new solar power tender has been launched in Brazil, aimed at fostering hybrid solar power projects that combine Photovoltaic (PV) and Concentrated Solar Power (CSP) technologies. This tender emphasizes the benefits of both technologies: PV's cost-effectiveness and CSP's ability to deliver dispatchable power. The maximum net output for grid delivery is capped at 100 MWe, ensuring a manageable and efficient plant design.

Our consortium, having been qualified for this tender, is preparing to submit a technical and financial proposal. To meet the requirements, we are conducting a techno-economic pre-feasibility analysis to determine the optimal configuration for a hybrid PV-CSP plant. This analysis includes location selection, plant configuration design, estimated annual performance, estimated costs, and expected base price for a Power Purchase Agreement (PPA) according to a specified tariff scheme.

The scope of the project involves selecting the optimal site, designing the plant's configuration, estimating annual performance, and calculating costs and Power Purchase Agreements (PPAs). The plant will have a maximum net output of 100 MWe, with at least 25% of the installed capacity coming from CSP to ensure dispatchability. A significant portion of the annual energy production, at least 30%, will be delivered during evening peak hours, between 17:00 and 24:00, to address high demand and maximize revenue. The project also considers an additional 20% remuneration for maintaining a continuous power output during peak periods.

The methodology includes thorough site selection, plant design, performance estimation, and cost analysis. The final proposal will outline the optimal dispatch strategy, energy production, and revenue projections based on different tariff periods. By combining PV and CSP technologies, this project aims to support Brazil's renewable energy goals while ensuring a stable and reliable power supply.

Abbreviations

AC / DC	Alternative Current / Direct Current
CAPEX	Capital Expenditure
CF	Capacity Factor
CSP	Concentrated Solar Power
DNI	Direct Nominal Irradiance
EH	Electric Heater
GCR	Ground Coverage Ratio
GHI	Global Horizontal Irradiance
HTF	Heat-Transfer Fluid
IRR	Internal Rate of Return
KPIs	Key Performance Indicators
LCOE	Levelized Cost of Electricity
LUC	Land Use Change
NPV	Net Present Value
OPEX	Operating Expenditure
PB	Power Block
PPA	Power Purchase Agreement
PV	Photovoltaics
SAM	System Advisor Model
SM	Solar Multiple
SPIs	Supportive Performance Indicators
STC	Standard Test Conditions
TES	Thermal Energy Storage
USD	United States Dollars

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1 INTRODUCTION

In this chapter we introduce our project, starting with a short background on Brazil and its relationship with PV and CSP technologies, followed by the presentation of our objective and limitations of this project, the KPIs and SPIs we will use for the analysis, our site selection processes, and we conclude with an explanation on our methodology.

1.1 Background

Brazil has the largest electricity market in Latin America, with a remarkable focus on renewable energy sources. The renewable energy sector dominates the Brazilian electricity matrix, accounting for a staggering 83% share, significantly surpassing the global average of approximately 25% (International Trade Administration, 2023). Notably, Brazil has over 39.8 GW of installed photovoltaic (PV) capacity, with distributed generation comprising 69% and centralized generation constituting the remaining 31%. Analyzing the Brazilian electricity mix reveals a diverse landscape, with hydropower emerging as the predominant source, contributing 48.3% of the total capacity. The country's extensive hydropower infrastructure, with an installed capacity exceeding 150 GW and a potential of 172 GW, underscores its significance in the energy portfolio (International Trade Administration, 2023). Solar PV follows closely, constituting 17.5% of the electricity mix and demonstrating rapid growth, with PV generation accounting for 10% of the total electricity supplied as of November 2023. Wind power also plays a substantial role, contributing 13% to the electricity mix with an installed capacity of approximately 29 GW. Additionally, Brazil utilizes other sources such as natural gas, biomass, biogas, oil, and other fossil fuels to diversify its energy mix and ensure energy security.

Brazil's energy landscape is a dynamic mix of renewable (hydropower, solar and wind) and non-renewable sources (biomass, oil, fossil fuel), with hydropower historically dominating electricity generation. Major hydroelectric plants have contributed substantially to the country's power supply, with hydropower accounting for over 60% of Brazil's electricity generation. However, this heavy reliance on hydropower makes Brazil vulnerable to fluctuations in rainfall, leading to occasional energy shortages during droughts. To mitigate this risk, Brazil has been diversifying its energy mix, with significant investments in wind and solar power. According to ABSOLAR and ANEEL, in 2023, solar PV represented 17.5% of the total power generation capacity with 39.8 GW of PV in operation. No CSP plant is in operation in Brazil for the moment.

At COP26 where Brazil announced a long-term objective to reach net-zero emissions by 2050 and a 50% carbon emissions reduction. Being one of the sunniest countries in the world gives Brazil the opportunity to become a leading nation in solar photovoltaic. The country ended 2023 in sixth place in the world ranking of photovoltaic solar energy generation, shows a report from the International Renewable Energy Agency (Irena 2024). Brazil's 2050 National Energy Plan (NEP 2050) outlines the importance of solar PV for Brazil's energy mix. Regarding utility-scale power plants, solar photovoltaics should reach an installed capacity of 90 GW in 2050 (NEP).

1.2 Objective

The objective of this project is to develop a hybrid PV-CSP solar power plant in Brazil, by conducting a technical and economic analysis to obtain the optimal dispatch strategy that combines

the best of both these technologies. With this, we aim to provide Brazil with cheap and dispatchable solar power.

The analysis consists of site selection, plant configuration design, annual performance estimation, cost, and PPA estimation.

1.3 Delimitations

The design of the plant comes with limitations and requirements. The maximum net power that can be delivered to the grid at any time is 100 MW_e, the CSP installed capacity should be a minimum of 25% of the installed PV capacity, and a minimum of 30% of the annual production, in kWh, should be injected into the grid during the evening peak hours, between 17h and 24h, to ensure an efficient and reliable energy supply during peak demand time. Continuously injecting firm power into the grid for 3 or more hours will earn the plant an additional remuneration of 20% over the base price during such hours. This motivates a focus on constant power output to maximize revenues. The final proposal will contain energy production share, revenue allocation, and breakdown by each tariff period, 0h-5h (night hours), 5h-17h (day hours), and 17h-24h (evening peak hours).

1.4 Key Performance Indicators (KPIs) & Supportive Performance Indicators (SPIs)

1.4.1 Economic KPIs

Power Purchase Agreement (PPA) (X Value)

The PPA price, or X value is the main economic factor of this project's analysis. It is one of the main components of the price bid since it presents the base tariff at which electricity is paid, during the off-peak period. A lower PPA price makes the plant more competitive in the market, so this project will optimize the plant's design for a minimum PPA. We obtain the value by setting the NPV to zero.

Weighted PPA

The Weighted PPA, or W_{PPA} , represents the real electricity price paid to the plant, being this way uses the given value α .

$$W_{PPA} = X \frac{E_{off\ peak} + \alpha E_{peak}}{E_{off\ peak} + E_{peak}}$$

Net Present Value (NPV)

The NPV is the sum of the discounted cost and income over the project's lifetime. It represents the global benefits/losses of the project: if the value is positive, the company is economically viable, if it is negative, the project is not profitable and the project shouldn't be executed.

The R value is the income from selling the electricity, depending on the hourly electricity production, E, the X value, and the hourly price factor α . The NPV is going to be set to zero in order to obtain our optimal X value.

The NPV is calculated through the following equation:

$$NPV = - \sum_{t=0}^{n_{con}-1} \frac{CAPEX}{n_{con}(1+IRR)^t} + \sum_{t=n_{con}}^{n_{con}+n_{op}-1} \frac{R(E, X, \alpha) - OPEX}{(1+IRR)^t}$$

1.4.2 Economic SPIs

Levelized cost of electricity (LCOE)

The LCOE represents the cost of producing one unit of electricity by dividing the lifetime costs by the energy production. It is calculated using the following formula:

$$LCOE = \frac{\alpha \cdot CAPEX + OPEX + \beta \cdot C_{dec}}{E}$$

With:

$$\alpha = \frac{(1+i)^n \cdot i}{(1+i)^n - 1} + k_{ins}$$

And,

$$\beta = \frac{i}{(1+i)^n - 1}$$

1.4.3 Technical SPIs

Capacity Factor

The capacity factor assesses the energy production of the plant, hybrid and individual when compared to their rated power, and it's given by the following equations:

$$CF = \frac{E}{8760 \times \text{Hybrid Plant Rated capacity}}$$

$$CF_{PV} = \frac{E_{PV}}{8760 \times \text{PV Plant Rated capacity}}$$

$$CF_{CSP} = \frac{E_{CSP}}{8760 \times \text{CSP Plant Rated capacity}}$$

Annual energy production

This value corresponds to the total energy production by the power plant over one year. And it is calculated for the hybrid plant (E), and individually for the PV plant (E_{PV}) and CSP plant (E_{CSP}).

The energy produced during evening peak hours

This value verifies that the dispatch strategy of the power plant obeys the limitation that stated that 30% of the annual total annual production of energy should occur during peak hours, and it is calculated using the following formula:

$$E_{peak} = \frac{\sum_{t=1}^{365} \sum_{h=17}^{24} E_{t,h}}{E}$$

1.5 Site Selection

For locating an optimal site for a hybrid PV-CSP solar powerplant in Brazil, the International Renewable Energy Agency's (IRENA) Global Atlas was used as the main tool. This tool offers comprehensive global data and maps for assessing any locations renewable energy potential, helping us pinpoint the regions that are most suitable for a solar powerplant in Brazil.

1.5.1 Solar Irradiation

Initially, our focus was directed towards analysing the solar irradiance of in Brazil. Using IRENA's tool, we determined the most relevant indices for our solar pre-feasibility assessment: the annual Global Horizontal Irradiance (GHI) for solar Photovoltaic (PV) production and the annual Direct Normal Irradiance (DNI) for Concentrated Solar Power (CSP) production. Our investigation revealed that the eastern regions of Brazil were particularly promising, as illustrated in Figure 1.

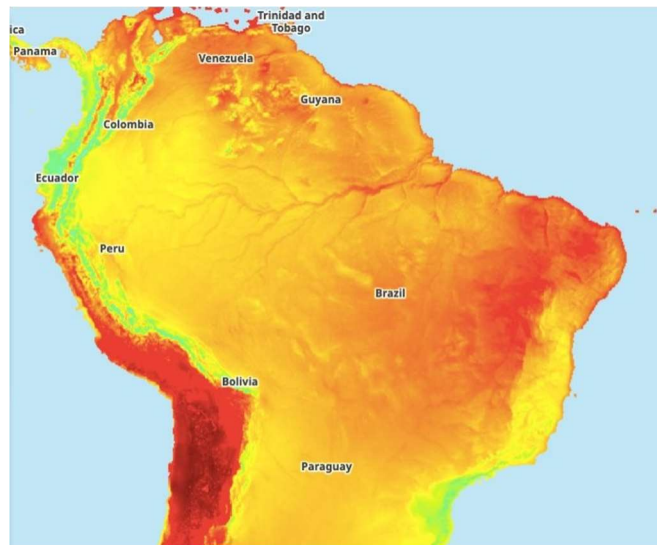


Figure 1: Solar Irradiance Map of Brazil

Analysis of the presented figure revealed three primary states showcasing the highest potential for solar energy generation: Bahia, Minas Gerais, and Pernambuco. Notably, these states boast average Global Horizontal Irradiance (GHI) and Direct Normal Irradiance (DNI) levels of 2100 kWh/m² and 1950 kWh/m², respectively. These values significantly surpass the global average. For comparison, Sweden's average annual GHI stands at 1000 kWh/m². These findings highlight the strong potential of these states as prime locations.

1.5.2. Integration with Transmission Network

In tandem with the solar irradiance assessment, an evaluation of Brazil's transmission network was conducted to ensure grid connectivity. Remarkably, the results of this analysis aligned with solar irradiance findings, affirming Bahia, Minas Gerais, and Pernambuco as well-covered areas within the transmission network, as illustrated in Figure 2.

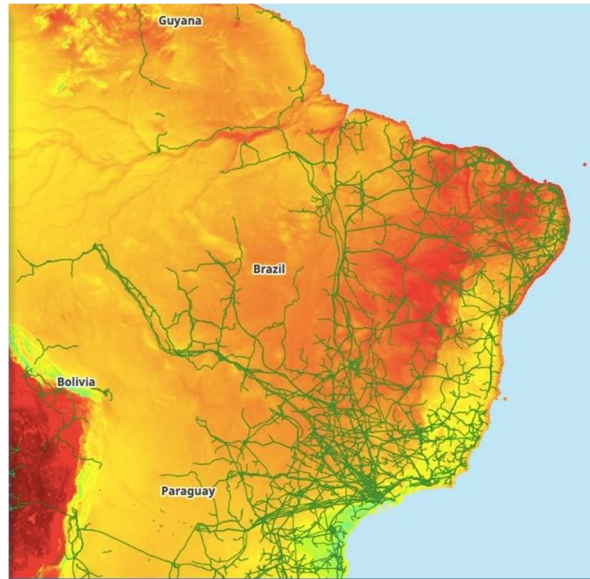


Figure 2: Transmission Network Map of Brazil

Furthermore, within the existing transmission network, there is considerable value in identifying the high voltage stations. The presence of these high-voltage locations, denoted by the red points on Figure 3, signifies areas where large-scale power plants can be strategically situated in close proximity. These locations serve as critical points within the transmission network, allowing for dispatching high voltage electricity over long distances.



Figure 3: High Voltage Transformers Map of Brazil (Red Points)

By positioning our powerplant close to these high voltage stations, the generated electricity can be integrated into the grid with minimum transmission losses, optimizing the output yield. Pairing the Irradiance and high voltage station maps in figure 4, we identify three potential sites near the towns

of Petrolina, Irecê, and Barreiras, all situated within the state of Bahia, as optimal locations for the integration of a hybrid PV-CSP solar power plant.

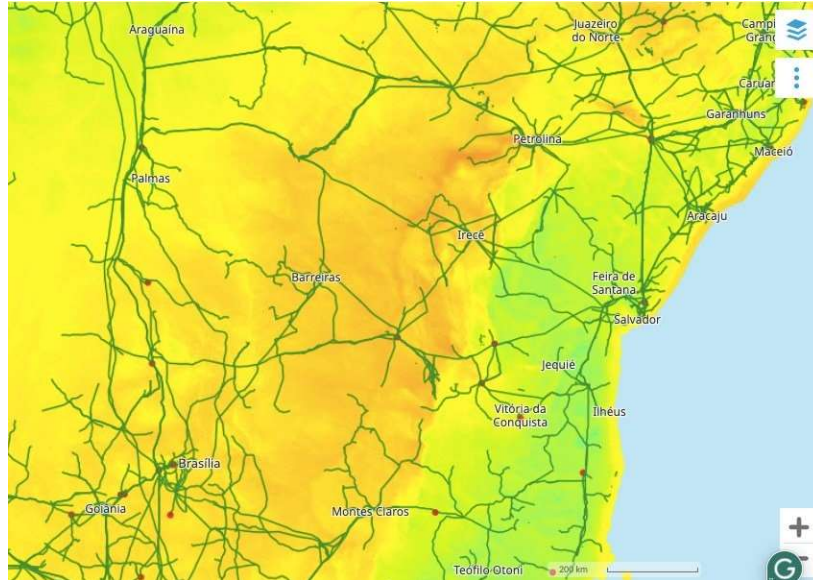


Figure 4: High Voltage Transformers Map layered on Solar Irradiance Map of Brazil

1.5.3. Additional Considerations

In addition to solar irradiance and transmission network coverage, a few other factors were considered to ensure sustainability as well as minimal social and ecological impact of the selected sites. These factors included site temperature range, population density (lighter areas in Figure 5), road networks, topography, and Land Use Change (LUC). To mitigate potential LUC, we committed to avoiding areas with significant ecological value or high agricultural productivity, thus minimizing disruption to existing land uses.

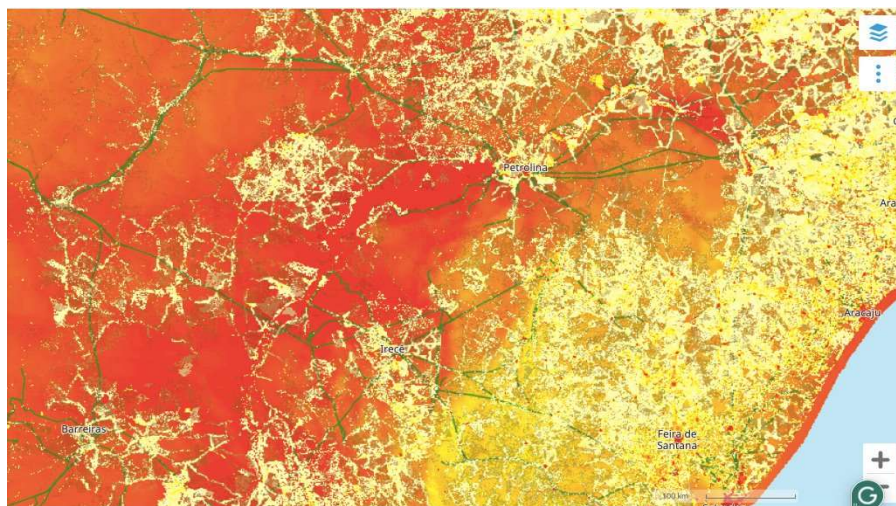


Figure 5: Population Density Map of Eastern Brazil

1.5.4. Conclusion

To conclude our analysis of potential sites for the placement of the hybrid PV-CSP solar powerplant, three coordinates were chosen and near the towns of Petrolina, Irecê, and Barreiras, all situated within the state of Bahia:

Table 1

Location 1: Petrolina (State of Bahia)

Coordinates	9°28'59.7"S 40°32'20.0"W
Annual GHI (kWh/m ²)	2150.6
Annual DNI (kWh/m ²)	1990.7
Closest substation	Subestação de Energia Elétrica CYMI Juazeiro III
Distance to closest substation (km)	3.3
Topography Type	Flat terrain
Site temperature range (°C)	24 – 28.4 (average: 26.6)
Distance to road (km)	0.98
Population density (ppl/km ²)	3.95
Land Use	Clear land, below grade 3

Table 2

Location 2: Irecê (State of Bahia)

Coordinates	11°19'25.4"S 41°40'58.9"W
Annual GHI (kWh/m ²)	2182.4
Annual DNI (kWh/m ²)	2002.74
Closest substation	Subestação Coelba - João Dourado
Distance to closest substation (km)	8
Topography Type	Flat terrain
Site temperature range (°C)	15 – 35 (average: 22.3)
Distance to road (km)	1.62
Population density (ppl/km ²)	2.88
Land Use	Clear land, below grade 3

Table 3

Location 3: Barreiras (State of Bahia)

Coordinates	12°02'13.2"S 44°40'16.3"W
Annual GHI (kWh/m ²)	2186.2
Annual DNI (kWh/m ²)	2003.1
Closest substation	Subestação Angical – Coelba
Distance to closest substation (km)	1.9
Topography Type	Flat terrain
Site temperature range (°C)	12.2 – 40.2 (average: 25.4)
Distance to road (km)	3.38
Population density (ppl/km ²)	4.03
Land Use	Clear land, below grade 3

In the evaluation of potential sites for the hybrid PV-CSP solar power plant, all three locations in the state of Bahia exhibited promising solar irradiation levels, with an average Global Horizontal Irradiance (GHI) of 2173.07 kWh/m² and an average Direct Normal Irradiance (DNI) of 1998.847 kWh/m². However, upon closer examination, the second location in Irecê stood out due to its lower site temperature range, ranging from 15°C to 35°C (average: 22.3°C), which is crucial for maximizing solar power plant output efficiency. Additionally, Irecê exhibits a lower population

density of 2.88 people/km² and a shorter distance to the road at 1.62 km, enhancing logistical accessibility and minimizing potential disruptions.

Considering these factors collectively, including solar irradiation levels, site temperature range, population density, and proximity to road and high voltage infrastructure, it was determined that the second location in Irecê offers the most favorable conditions for the implementation of the hybrid PV-CSP solar power plant project.

1.6 Methodology

The optimization goal is to design the PV and CSP plant enabling a minimum PPA price (X price). The optimization of our plant is composed of two main phases. The first phase aims at optimizing the PV plant and the CSP plant separately. These first optimizations are based on literature review and on simulations on SAM: the outputs of the first phase are various design of PV plants and CSP plants. From this phase we selected one of the best PV designs and got an idea of the optimized design for the CSP part. The second phase focuses on the optimization of the dispatch strategy for the hybrid plant, enabling us to continue optimizing the CSP plant. To do so, we have created a code that will optimize the dispatch of the CSP energy according to the PV production and the hourly prices (α_{factor}). The code output will be the dispatch strategy for the CSP part. This dispatch will help us finalize the design of the CSP plant by modifying some key technical parameters to maximize the revenue. **At the end of these two phases, the objective is to identify the configurations of PV and CSP enabling to have the lower PPA price.** The Figure 6 represents this methodology.

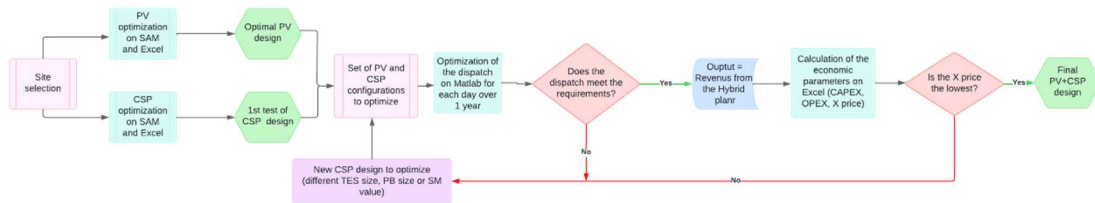


Figure 6: Methodology for the optimization process

2 PV DESIGN AND OPTIMIZATION

2.1 Methodology

In analysing and optimizing the PV component of the hybrid powerplant, an investigation was conducted to analyse the influence of various parameters on the PV output. For this analysis, we investigated different panel technologies, inverter configurations, installed DC capacities ranging from 100 to 225 MW, tilt angles (fixed tilt and tracking), and DC to AC ratios (1.25 and 1.4). Each aspect was examined to understand its impact on the system performance.

Additionally, a separate cost analysis of the PV system was conducted to determine how different configurations of the same components could lead to the lowest X price. This analysis aimed to understand the cost-effectiveness of various configurations by testing different combinations of DC capacity, DC to AC ratio, tracking systems, tilt angle. Through this approach, we aimed to optimize the PV output and bonus hours, essential for calculating the financial metrics such Capital Expenditure (Capex), Operational Expenditure (OPEX), Revenues, and ultimately, the X price. Figure 7 is the methodology flow chart of our process.

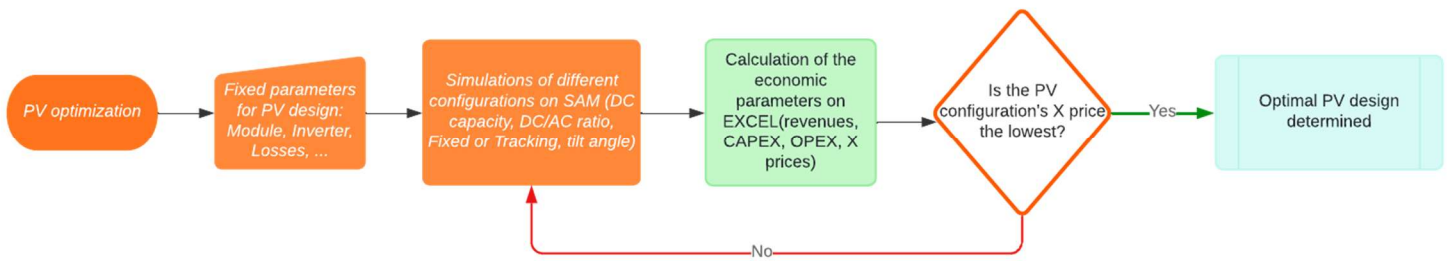


Figure 7: Methodology for the

PV optimization

2.2 Panel and Inverter Technologies

In choosing a panel technology for the specifications of our hybrid PV-CSP plant, our primary goal was to maximize output yield. Building on our knowledge from the large-scale solar course, we leaned towards mono-CSI and multi-CSI technologies, known for their higher efficiencies, versatility, and widespread use in commercial scales. While acknowledging that Mono-Csi panels tend to be more expensive, we recognized their higher efficiency, which could potentially offset the initial investment costs over the long term.

Using the SAM directory, we filtered a few panel technologies based on three key criteria: high output yield at Standard Test Conditions (STC) of 500-600 Wdc, along with high efficiency and low temperature coefficients. Our selection was as follows:

Table 4. Panel Selection Models

Model	Type	Pmp at STC (Wdc)	Efficiency (%)	Temperature coefficient (%/°C)
Tangshan Haitai New Energy Technology Co. HTM 660MH8-66	Mono-CSI	660.675	21.24	-0.345
WAAREE Energies Bifacial model Bi-68-660	Mono-CSI	661.035	22.03	-0.337

Astronenergy Astro6 Bifacial CHSM66M (DG)/F-BH	Mono-CSi	660	21.2	0.34
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Upon examination of the existing technologies, we saw that mono-CSi models were much higher in efficiencies and outputs compared to multi-CSi and thin film models. Additionally, taking Brazil's hot climate into account, the temperature coefficient of the models is of great importance. As a result, we singled out the WAAREE Energies Bifacial model Bi-68-660 as the most promising choice for our project; with the highest yield of 661.035 Wdc, 22.03% efficiency, and lowest temperature coefficient of -0.337.

For the inverter selection, we have selected Sichuan Clou Energy Electric Co. in the model NEPCS-2300 [800V]. With an efficiency of 98.566%, it allows for a seamless integration with the panel technologies. The maximum input (DC) and output (AC) powers of the inverter are 2335050 and 2299960 MW, respectively, which allows for flexibility when choosing the plant design and size, while also allowing the reduction of initial and maintenance costs by allowing a smaller number of inverters when compared to other models with less capacity.

2.3 Different PV Scenarios

Following the panel and inverter technologies selection, we proceeded to run initial plant simulations with various DC field sizes, DC-AC ratios and tracking systems, and for each, we did an initial basic comparative economic analysis by calculating the X price solely for the PV design, which will be used to decide on the final PV design.

The orientation of the PV array was constant throughout all the scenarios with a fixed tracking system with an azimuth angle of 0°, and of the systems with tracking, a 1-Axis tracking system with 45° of rotation limit. The GCR value used was 0.3. The module orientation is Portrait, the number of modules alongside per row is 2, and along the bottom of the row we have 31 modules. The losses considered in SAM were the default ones given by the program which can be seen in the following Table:

Table 5: Solar Field Losses considered in SAM.

Average annual soiling	Module Mismatch	Diodes and Connections	DC Wiring	Tracking Error	AC Wiring	Transmission Loss
5%	2%	0.5%	2%	1%	1%	1%

To start, with the different analysed scenarios with a DC-AC ratio of 1.25 we obtained the following results:

Table 6: Simulation results for an AC-DC ratio of 1.25.

Scenario	$P_{PV,des}$ (MWdc)	$P_{PV,ac}$ (MWac)	Tracking System	Tilt Angle	Annual Production (GWhac)	X price (USD/MWh)
PV1	150	120	Fixed	11.32°	241	67.23

PV2	175	140	Fixed	11.32°	266.2	66.38
PV3	200	161	Fixed	11.32°	283.1	66.26
PV4	225	179	Fixed	11.32°	295.8	71.18
PV5	150	120	1-Axis	-	281.3	64.42
PV6	175	140	1-Axis	-	301.8	66.4
PV7	200	161	1-Axis	-	314.1	70.92

Seeing as the scenario PV3 is the scenario that results in the lowest X price with a fixed tracking system we did a parametric analysis on SAM to obtain the tilt angle that yields the best results. The process consisted of using the parametric function in SAM and centring the range of values evaluated around the max value per round while reducing the step in the values analysed. The final result showed that the latitude was the optimal tilt angle, 11.32°. As a result, with a DC-AC ratio of 1.25, the best scenario is PV5, despite the increased cost of having a 1-Axis Tracking system, the X price we can obtain is the lowest and so for the optimal scenario for this ratio.

For a DC-AC ratio of 1.4 we obtain the following results:

Table 7: Simulation results for an AC-DC ratio of 1.4.

Scenario	$P_{pv,des}$ (MWdc)	$P_{pv,ac}$ (MWac)	Tracking System	Tilt Angle	Annual Production (GWhac)	X price (USD/MWh)
PV1*	150	108	Fixed	11.32°	241	67.41
PV2*	175	124	Fixed	11.32°	266.2	66.38
PV3*	200	143	Fixed	11.32°	283.1	68.26
PV4*	225	161	Fixed	11.32°	295.8	71.18
PV5*	150	108	1-Axis	-	281.3	64.42
PV6*	175	124	1-Axis	-	301.8	66.38
PV7*	200	143	1-Axis	-	314.1	70.92

Following this analysis, we followed the same procedure as before to analyse the effect of the tilt angle and what value would be the optimal one, the result obtained was the same, as the optimal tilt angle is the latitude. So, the optimal scenario using a 1.4 DC-AC ratio is PV5*.

When comparing the results with both tested ratios we can see that, due to their similarity, the results are almost unchanged, leading us to conclude that changing the ratio within our chosen range leads to almost no change in the results.

When analysing all the scenarios, we can see that the one that yields the lowest X price is PV5 which will be further presented in the following section. A visualization of the evolution of X prices can be seen in Appendix A.

2.4 PV field final design

When analysing all the scenarios, we can see that the one that yields the lowest X price is PV5 which will be further presented in the following section (below Table 8 and Table 9).

Module reference	Inverter reference	DC capacity	DC-AC ratio	AC capacity	Tracking system
WAAREE Energies Bifacial model Bi-68-660	Sichuan Clou Energy Electric Co. NEPCS-2300 [800V]	150 MW	1.25	120 MW	1-Axis With 45° rotation limit

Table 8: Design parameters of the selected PV plant

Annual production	CAPEX	OPEX	Estimated X price
281.3 MWh	267.4 MUSD	2.52 MUSD	64.42 USD/MWh

Table 9: Estimated annual performance and cost of the selected PV design

The final design of the plant occupies a land area of 680 760 m^2 . The final configuration consists of 226 920 modules per subarray, with 31 modules per string, and 7 320 in parallel. The annual profile of this configuration in comparison with the GHI can be seen in Figure 8. There we can see that the curve follows the expected shape of a 1-axis tracking system, relatively smooth following the GHI shape and a bit larger, with excess production in relation to the GHI in the hours with more sun.

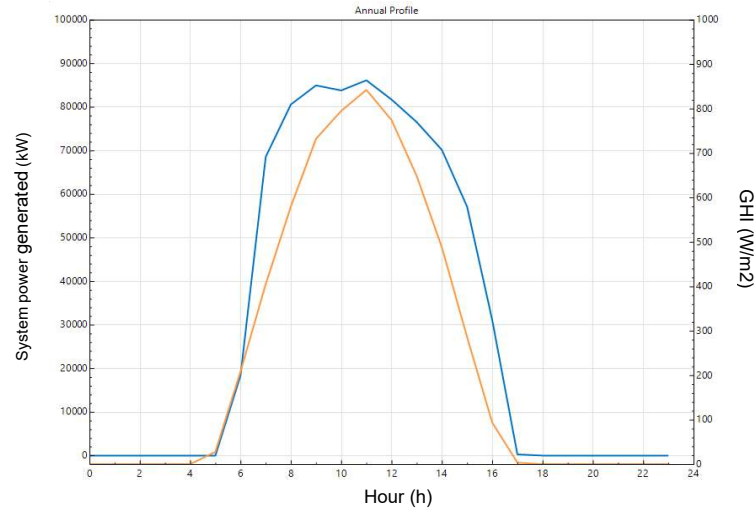


Figure 8: Annual solar production and GHI

The X price presented in Table 9 is not the final PPA price used for the plant. It is the price that was estimated to compare the economic aspects of the various PV designs considered. Since the total plant configuration doesn't include an electric heater, all the PV produced is directly sold to the grid, so an approximate X value for the PV part only is easily calculated compared to other configurations that include an electric heater. So, this calculation enables the choice of the optimal PV design regarding, not only, technical consideration, but also financial.

This PV configuration was then used to design the CSP plant and to optimize its dispatch according to the tender requirements and the alpha factor. The methodology is presented in the following part of this report.

3.1 Methodology

Several choices have been made regarding the CSP part of the hybrid plant. First was the choice of the technology. Having only one technology is more cost-effective than using both parabolic troughs and tower. (Guccione, 2024) Tower technology has been chosen over parabolic troughs for two main reasons: its higher concentration ratio, which leads to higher temperatures, and the better efficiency of the power block. The second design choice made for the CSP is to integrate the thermal energy storage directly in the system. This allows the use of the same Heat-Transfer Fluid (HTF) in the storage and the system, which helps to reduce the overall cost of the system and have a simpler design.

In order to assess the best CSP design, several parameters are taken into account. It has been decided that the following parameters would be the main ones for the optimization process: the net power output of the Power Block (PB), the Solar Multiple (SM) and the size of the Thermal Energy Storage (TES).

The parameters linked to the tower and the heliostat field are dependent of the previous parameters. Their values were calculated for each configuration using the optimization tool in SAM. Other parameters were fixed and used to form a basis for the comparison.

The same production strategy has been chosen for all configurations, which is to store as much as possible in the TES during the day and produce electricity in the evening when the PV part of the plant stops producing. This strategy is not the final one that will be used for the final optimization. It only helps comparing the CSP part independently by having an easy strategy to set. The optimization of the dispatch strategy is presented in part 4 *Method for energy dispatch* of this report.

In the end, the analysis has been conducted by implementing the different configurations of parameters in SAM, then extracting the hourly production and post-processing the data to get the X value necessary for each configuration. This X value is the parameter used for comparison of scenarios, as it represents the necessary selling price to be cost-effective, and therefore gives a good idea of the balance between the investment and the expected energy production.

3.2 Net Power Output

The first variable chosen to design the CSP part is the Net Power Output of the Power Block. SAM only allows the input of the gross output and applies a gross-to-net efficiency to the value based on the gross input, the solar multiple and the TES size. Therefore, all values of Net Output will be applied a coefficient before their implementation in SAM. The optimal power must minimize the overall costs of production while allowing enough energy production. The minimal power chosen here corresponds to the requirement of having 25% of the PV capacity in CSP. As the best PV scenario uses a capacity of 150 MW DC, the first estimation for the CSP is 40 MW. Three other powers were tested: 60 MW, 80 MW and 100 MW.

3.3 Solar Multiple

The second parameter is the solar multiple. It directly affects the total thermal energy reaching the receiver and therefore the possible stored energy of the plant.

It has been decided to start the configurations with a solar multiple of 1.2 and go up to 1.8 with an interval of 0.2.

3.4 Thermal Energy Storage size

The third and last parameter is the size of the Thermal Energy Storage (TES). It has been decided to choose the values in hours. 1 hour of TES represents 1 hour of power block running at nominal capacity. The initial choice for the storage is 7 hours to be able to run all over the peak hours period of 7 hours. Two other values are tested: 8 and 9 hours.

3.5 Control Parameters

All other parameters are kept constant for every configuration. The values and reasons for them are available in the following table:

Parameter	Value	Reason
Design Point DNI	1127.93 W/m ²	Maximum DNI in the weather file
Power Block thermal efficiency	0.38	Values from 0.35 to 0.38 for a usual Rankine cycle power block (Muñoz et al., 2022)
Heat-Transfer Fluid (HTF) temperature range	Hot: 574°C Cold: 290°C	-Classical range for molten salts between 270 and 590°C (Trevisan, 2023) -Default values in SAM

Table 10: Control parameters for CSP optimization

3.6 Results

The results of the optimization based on the described scenarios are provided in the following table. Only the best scenario (i.e. with the lower X value) for each net power output is presented, all other scenarios are available in the Appendix B.

Net Power Output	Solar Multiple	TES size	X value (USD/MWh)
40	1.8	9h	104.23
60	1.8	9h	88.74
80	1.8	9h	80.33
100	1.8	9h	81.60

Table 11: CSP optimization results for each power output

For a given power output, the X value decreases as the solar multiple and the TES size increase. This shows that the higher energy production overcomes the increase in the cost of the storage and heliostat field.

The best scenario of all is the one with a net power output of 80 MW. This means that for an output of 100 MW and over, the increase in energy production is not enough to compensate for the higher cost of the investment. However, this configuration is the best only in the specific case of CSP being studied independently, without the hybridization with PV. A new optimization will be conducted in the next steps of this report.

4 METHOD FOR ENERGY DISPATCH

4.1 Methodology

Several choices have been made for the dispatch strategy. The main characteristic of our strategy is that we chose to rely only on the base components of the hybrid PV-CSP plant (PV field, inverters, heliostat field, tower, power block and TES). This means that we did not take into account any Battery storage system or electric heater. The idea behind these choices is to avoid supplementary costs and try to reach the best energy dispatch for the base plant.

This strategy implies that any electricity generated by the PV part of the plant during the day is delivered and sold to the grid. The only constraint is that the PV does not produce if the selling price is negative, which would mean paying to deliver electricity. In such a case, PV production is simply curtailed. This PV generation is therefore our daily base and the generation through the power block is adapted to this base load.

The dispatch of the power block generation is thought of as daily planning based on the selling prices. Based on the irradiation data, the total daily amount of energy harnessed via the CSP system and stored in the TES is calculated. Then, for a period of time going from sunrise to the next sunrise (24 hours, with both day and night), the selling prices are sorted thanks to the alpha factor. Eventually, according to the different constraints, the maximum power possible is associated with each hour starting from the best-selling price and until all energy harnessed during the day is distributed.

Several constraints have been implemented to fit the bid requirements and the feasibility of the dispatch. First, to make sure that 30% of the final generation happened in peak hours, we designed our strategy so that this constraint is met every day. Secondly, we made sure that the power block could not run with a load lower than 30% of its maximum capacity. This is an average value for this kind of power cycle.

This strategy focuses on having the best gross revenues for each day, by producing when prices are the highest. The bonus for continuous firm production has not been taken into account for the dispatch. It is possible that the dispatch will make the bonus happen, but this is not the primary goal. In fact, the best hours in terms of selling price can be split evenly over the day and not be consecutive.

A second approach for the dispatch, based on linear optimization, has been studied during this project. The detailed method is available in Appendix D. However, due to the higher complexity of this method and the lack of time, it has been decided to leave it as a recommendation for future work.

4.2 Code for the dispatch

We aim to develop a code capable of managing this hybrid PV-CSP Production. To simplify the model, we have considered maximum PV output (AC) to be capped around 100MW and the dispatch to the grid is not controlled. The optimal dispatch for the CSP is decided on a day-to-day operation, the decision variable for CSP dispatch is determined by sorting the time-of-the-day differential (TOD is alpha factor) from highest to lowest and dispatching the accumulated thermal

energy as per the sorting. The flowchart explains the dispatch strategy in detail and the full code is available in Appendix C.

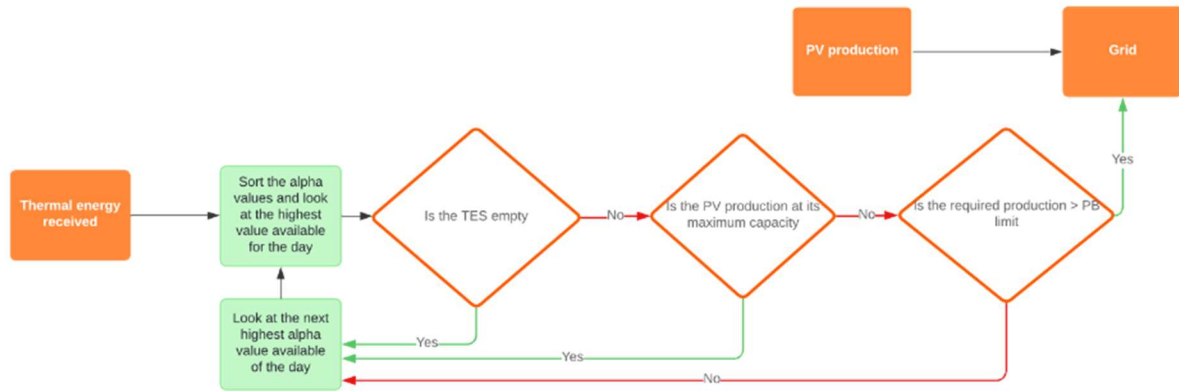


Figure 9: Flow chart of the dispatch strategy

A few assumptions and modelling choices are made in the code to simplify the calculation and the implementation of the strategy. The first one is that the TES is considered empty at the beginning and at the end of each day. This means that all the thermal energy harnessed during a day is stored and sent to the PB during the same day. The second one is that no thermal losses are considered in the TES over the timespan of day. This allowed an easier distribution of production, even if the final generation results are therefore slightly higher than they would be in reality. The last modelling choice is that the sunrise has been fixed at 6 AM every day of the year. We indeed chose to define each day as a period of 24 hours from 6 AM to 5 AM the next day, the next sunrise. In reality, the sunrise time goes from 6 in summer to 7:30 in winter. However, this modelling choice does not change results as the whole daytime is always covered.

5 RESULTS & ANALYSIS

This section presents the PV-CSP design that yields the best results and is therefore selected for the compliant bid.

5.1 The selected design

A power plant composed of 150 MWDC PV and 80 MWe power block CSP has been designed. It enables us to produce a yearly 459.5 GWh of electricity, with a **capacity factor of 52%**.

Table X displays the main characteristics of the power plant on the technical side. The values mainly result from the optimization presented in section 2 and 3. Some parameters such as the storage size or the size of the power block could not be optimized for the CSP independently as it depended on the PV production and the dispatch strategy. As a result, these parameters have been optimized when calculating the revenue from the dispatch in section 5. The complete table with all the technical parameters, such as the module chosen and the number of arrays, is presented in Appendix E.

PV		CSP		Hybrid Plant	
DC Capacity	150 MW	Power Block DC capacity	80 MW	Grid capacity	100 MW _{ac}
DC-AC ratio	1.25	Solar Multiple	1.8		
Mounting system	1-Axis Tracking	TES size	9 hours	Global annual production	459.5 GWh
Azimuth	North (0°)	HTF	Molten Salt		
Tilt angle	-	Solar Field Area	3,796,685 m ²		
Land Area	680,760 m ²	Tower Height	154.2 m		
Capacity factor	27 %	Capacity factor	25 %	Capacity Factor	52 %

Table 12: Parameters of the technical bid

The capacity factor value of the PV plant corresponds to the usual one for a 1-Axis tracking system (Boretti A. 2018), same for the CSP plant alone (Ma Zhiwen and al., 2012). The global plant capacity factor is above the ones of PV and CSP alone as expected: as the overall AC capacity is 100 MW and the PV and CSP do not produce simultaneously, the balance between production and maximum production is good, enabling a capacity factor of 52%.

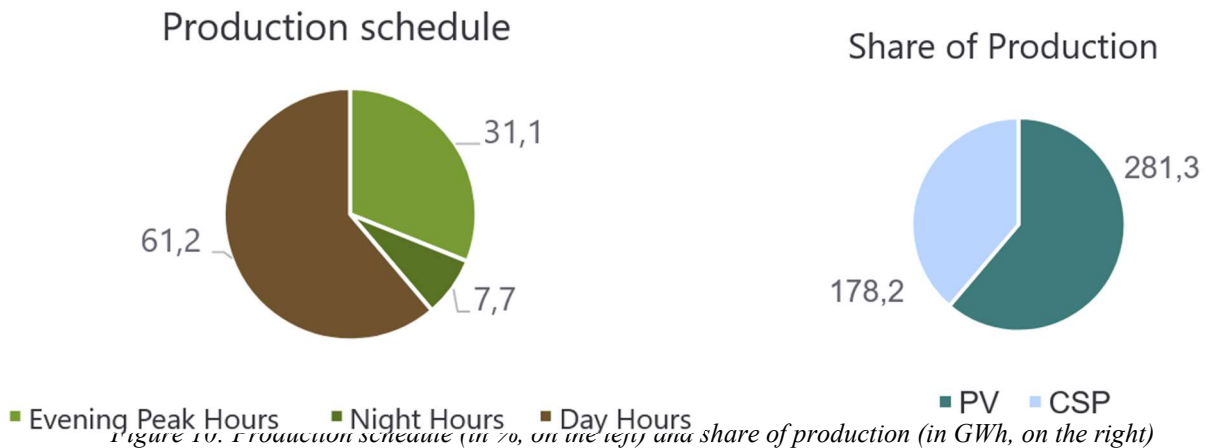
5.2 PPA price

The PPA price was obtained by targeting a NPV equal to 0 at the end of the lifetime of the hybrid plant, that is to say 32 years (2 years of construction and 30 years of operation). After having calculated the revenue from the dispatch code, the bonus revenue from the constant production and the CAPEX and OPEX, we used Excel to obtain our PPA price. With our selected design, our team proposes a **PPA price of 74.49 USD/MWh**. The details of the financial analysis are presented in the section 5.4.

5.3 Technical Analysis

The following pie chart represents the breakdown annual production regarding the different schedules of the day: the day hours (5am-5pm) represents 61.2%, the evening peak hours (5pm-12pm) 31.1% and the night hours 7.7%. **Thus, the requirement of having at least 30% of the global production during the evening peak hours is fulfilled.**

In addition, as expected, the PV panels produced a major part of the energy, 281.3 GWh per year, representing 61% of the total share. We can see that the CSP is meant to produced only for the highest paid hours, which are the peak hours: the strategy presented in section 4 is effective regarding the 30% peak hours requirement. The analysis of the revenue is presented in the following section.



Based on the dispatch strategy established previously, the production profile for typical days can be extracted. The PV generation is fixed during daytime in both cases, curtailed and capped to 100MW in December (summer) and barely reaching the maximum in June (winter). The CSP production happens at peak hours and nighttime, when the prices are higher. This is visible in Figure 11 on the alpha factor curve.



Figure 11: Production profile for CSP and PV for two typical days (December 21st and June 21st)

5.4 Financial Analysis

In order to perform the economic analysis, the CAPEX and OPEX for this hybrid plant were calculated using the Excel file provided, Generic CAPEX OPEX model CSP-PV MJ2500.xlsx. The Total CAPEX obtained was 631.4 MUS\$D, with a direct CAPEX of 522 MUS\$D and 109.4 MUS\$D of indirect CAPEX. The breakdown is in Figure 12.

Here we can see that 51% of the total CAPEX corresponds to the CSP part, while only 22% is for the PV part. The biggest investment is made in the CSP solar field, followed by Engineering costs and the CSP BOP (Balance of Plant).

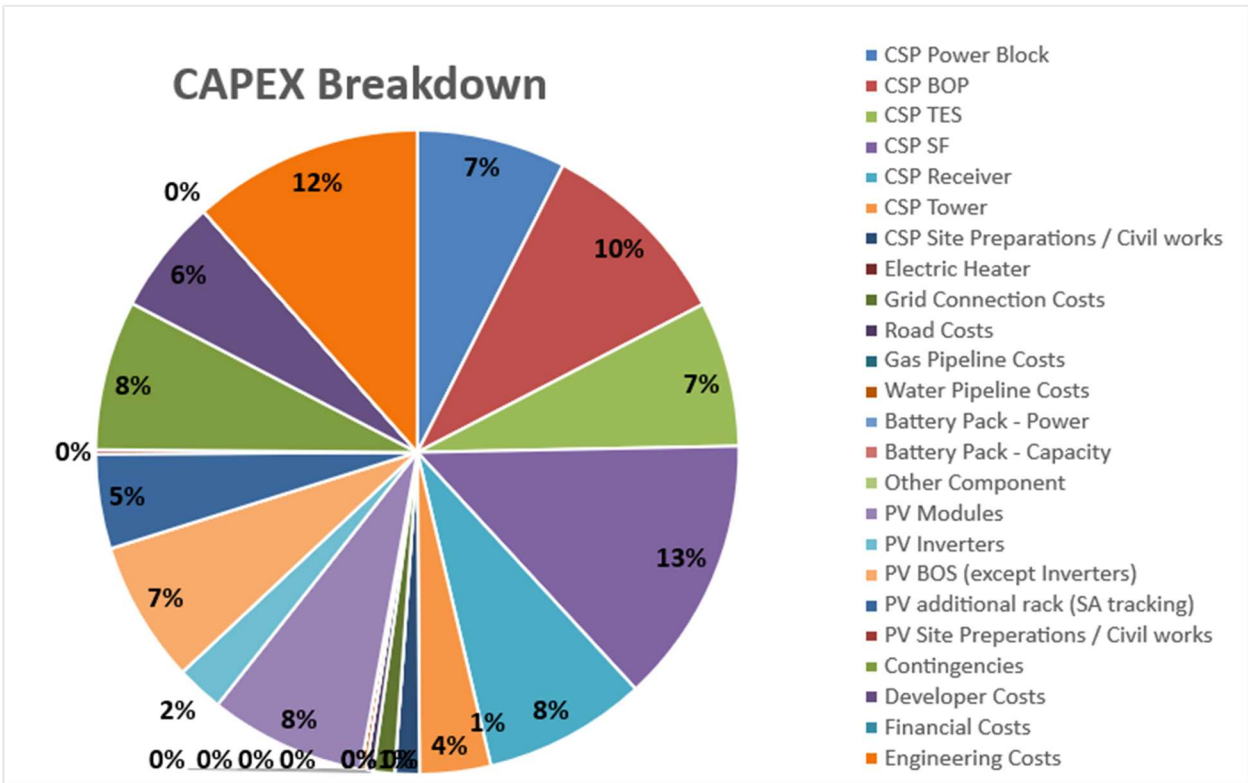


Figure 12: CAPEX breakdown

The annual OPEX obtained was 6.88 MUSD, and in the breakdown that follows we can see most of the OPEX is in relation to the CSP part of the plant with 64%, and the PV with 36%. For CSP the highest operational costs are for the CSP Plant Operations Labor and CSP Electricity Use.

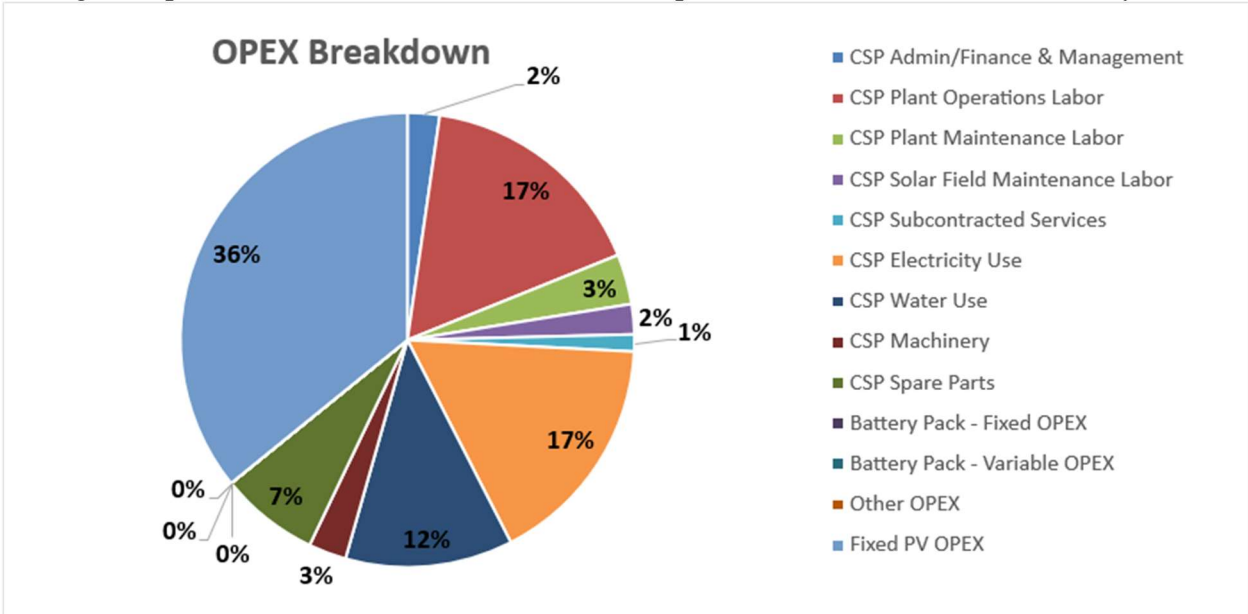


Figure 13: OPEX breakdown

The annual revenue obtained, considering the X value, the annual production and the bonus is 36.36 MUSD. 4.18 MUSD out of this total is due to the bonus which is verified in 35% of the time.

For the calculation of the final X price, the NPV was put to 0, so the payback period is the lifetime. The cash flow over the lifetime of this project used for this calculation is presented in Appendix F.

In the following graph, we can see the share of the total revenue by tariff schedule. Here we can see that most of the revenue is obtained during the evening peak hours in which 31.1% of the total generation is produced. Despite most of the electricity being generated during day hours, 41.18% of the total revenue is generated during these hours, this is due to lower alpha values and less stable generation resulting in less bonus during these hours compared to the evening peak hours. This high share revenue during the evening hours confirms the efficiency of the CSP dispatch strategy leading to a competitive PPA price.

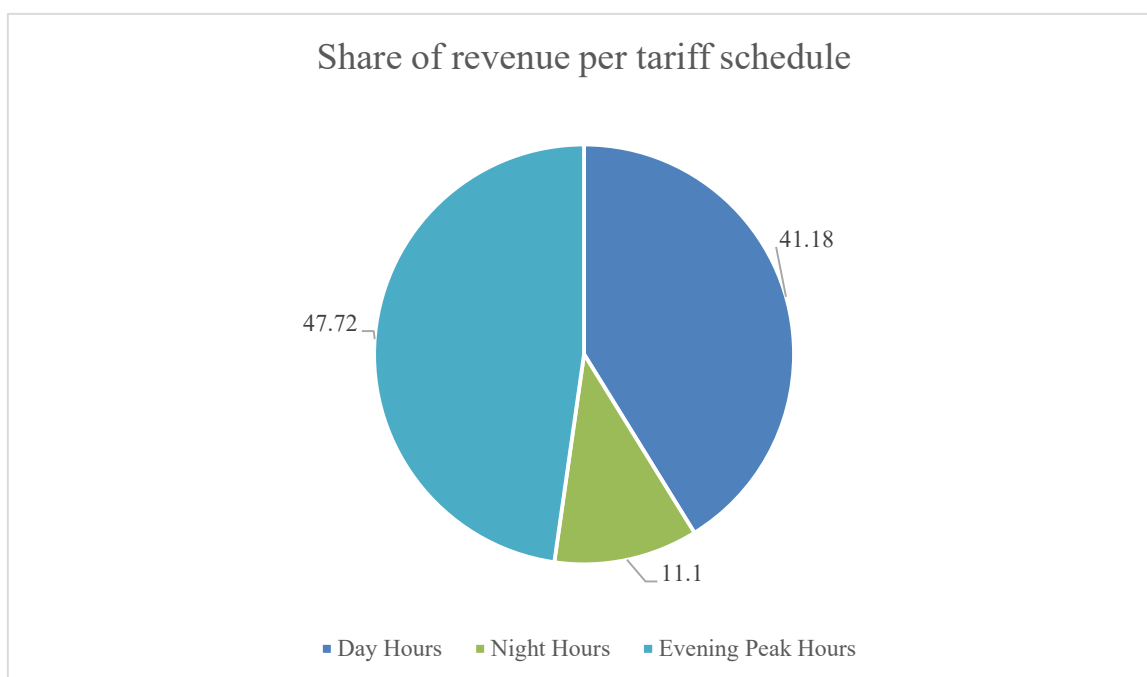


Figure 14: Share of revenue per tariff schedule (in %)

As part of our analysis, we calculated the Levelized Cost of Electricity (LCOE) for this project, which came out to **143 USD/MWh**.

This is higher than the PPA as expected, and can be attributed to several factors, including the lower capacities installed in the plant and the absence of an electric heater, which resulted in 3.84% of electricity being clipped (from PV). Despite this, when we compare our LCOE to those of hybrid PV and CSP projects in Chile (Parrado, C. *et al.*, 2016) as shown in Figure 14, our value falls within the range observed in 2016. This indicates that while our LCOE is high, it is not out of line with similar projects. Our primary focus was on optimizing revenue rather than production, which explains the low PPA price that includes the revenues in its calculation, but higher LCOE that only takes into account the costs compared to the annual production.

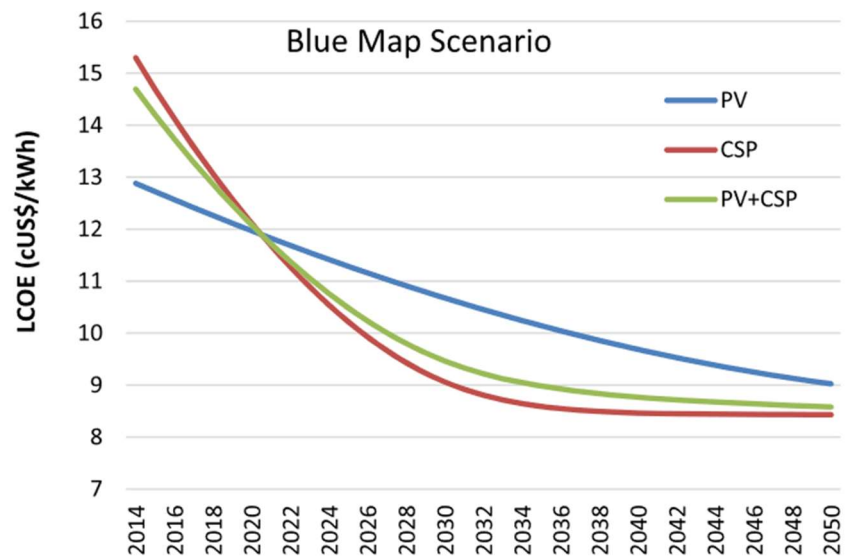


Figure 15: Evolution of the LCOE price of PV, CSP and hybrid installations in Chile (Parrado, C. et al., 2016)

6.1 *Environmental impacts assessment*

Our hybrid PV and CSP plant will contribute positively to SDG 13 “Climate action”, SDG 7 “Affordable and clean energy”, SDG 8 “Decent work and economic growth” and SDG 3 “Good health and well-being”.

Indeed, on the one hand, this plant will enable Brazil to develop its renewable energy production, aligning with Brazil COP26 long-term objective to reach net-zero emissions by 2050 and a 50% carbon emissions reduction. The total amount of emissions avoided with this project can be calculated compared to Brazil CO₂ emission factor for its electricity production: in 2023, the emission factor of the global electricity mix was 86.64 gCO₂eq/kWh (IEA, 2023) and for solar PV and CSP we consider 40 gCO₂eq/kWh (NREL).

The total emissions avoided are 21 140 tons of CO₂ per year, which corresponds to 9,200 flights from Stockholm to Rio de Janeiro for one person.

This reduction of emission contributes to the SDG 13 and to the SD7 and SDG 3 as the population of Irêce will benefit from clean energy.

However, we did not consider the impact of our plant on the local biodiversity. This large-scale solar power plant can have significant impacts on the country's rich biodiversity and environment. The construction of the facility will surely involve clearing and grading of land, which can lead to habitat loss and degradation for many plant and animal species (Gardenio Diogo Pimentel da Silva, 2019). This can result in species mortality, displacement, and declines in species richness and density (OCDE, 2024). Additionally, the ongoing operation of solar plants, such as the use of herbicides and frequent mowing, can further degrade habitats and disrupt ecosystem functions (OCDE, 2024). To address these challenges, solar projects should be carefully planned and implemented with measures to minimize environmental impacts, such as co-locating solar with agriculture, protecting sensitive habitats, and implementing biodiversity monitoring and conservation programs. This aligns with Sustainable Development Goal 15, which aims to "protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and biodiversity loss."

Lastly, the development of the solar energy contributes to the diversification of the Brazilian electricity mix: it will enable Brazil to minimize its environmental impacts by reducing its reliance on hydropower. Indeed, climate change, manifested in unpredictable precipitation patterns, water scarcity, and extreme weather events, poses challenges to hydropower in this country. Our project contributes to the reduction of the vulnerability to climate-related disruptions, minimizes competition for water resources, enhances resilience against extreme weather, and aligns with sustainable development goals by mitigating environmental impacts associated with large-scale hydropower.

6.2 *Economic impacts assessment*

On the other hand, this plant will have a positive impact on Irêce local economy. The population of this region will benefit from an affordable source of energy. Moreover, the construction of the plant and its flow of materials and workforce will generate revenues for the local communities and companies. During the lifetime of the plant, an operation team will also be needed; long-term jobs will thus be created. All these elements contribute to the SDG 8.

7 DISCUSSIONS & CONCLUSIONS

This section aims at analysing several limitations that have been identified throughout the project and discusses some potential further improvement.

7.1 Discussions

▪ Methodology

For this project, we analyzed the PV and CSP parts separately and created a code to optimize how we dispatch the power. However, we did not optimize the technical details of the CSP plant in the code but tested different configurations. This strategy is limited as it takes time to run the code for the various configurations studied. The configurations were selected after a first simplified optimization using SAM as presented in section 3. However, the selection was made according to a quick estimation of the PPA price that did not consider the future dispatch. This allowed us to focus on the energy output but left some technical aspects unexplored regarding the CSP and TES design.

• Site Selection

Our site selection was quite basic, as we wanted to concentrate on optimizing the plant's performance. Ideally, we would also consider grid capacity, environmental impacts, and the presence of rare species. A more detailed approach would be necessary for complete and sustainable site selection.

• PV Design

In designing the PV system, we used fixed parameters like system losses and GCR. We calculated shadowing effects using SAM but didn't optimize them. Future work could focus on reducing shadow losses by adjusting row spacing or improving the shadowing calculations by using the other methods proposed on SAM, which could increase energy production.

• Dispatch Strategy and model limitations

We optimized the dispatch strategy using the alpha factor but didn't consider the bonus hours, which could increase revenue. Also, our strategy only looked at the minimum production needed for the power block. Starting and stopping the power block for just one hour might not be practical. Moreover, the current model does not account for excess PV generation which if considered would require an electric heater and a dispatch strategy which isn't considered in the project. What's more, the alpha factors for one day ahead are not considered, which is due to the decision variable used (sorting alpha for one day and dispatch based on it). Regarding all these constraints, we started looking at a more complex and complete algorithm created by Juan Maz Zappater (2023). We started implementing it at the beginning of the project but due to the project's short time frame, we continued with the code and dispatch strategy presented in section 4. However, so as to highlight the work done on this, the global ideas of the second code and its partial implementation are presented in Appendix D.

• TES Losses

We did not account for Thermal Energy Storage (TES) losses in this project. According to Ma et al. (2012), these losses are less than 1% per hour, so their impact is minor. However, future analyses should include TES losses for accuracy.

• Cost Assumptions

The cost data were taken from a provided Excel file and might not be adjusted for Brazil. However, we assumed the same land cost across all of Brazil, which is unrealistic. Future studies should consider regional variations in land costs.

Overall, we were aware of the assumptions made during this project to simplify our optimization to meet the deadlines and the project requirements. The results were analyzed compared to the expected results from the courses and some literature to present an accurate and realistic bid.

7.2 Conclusion

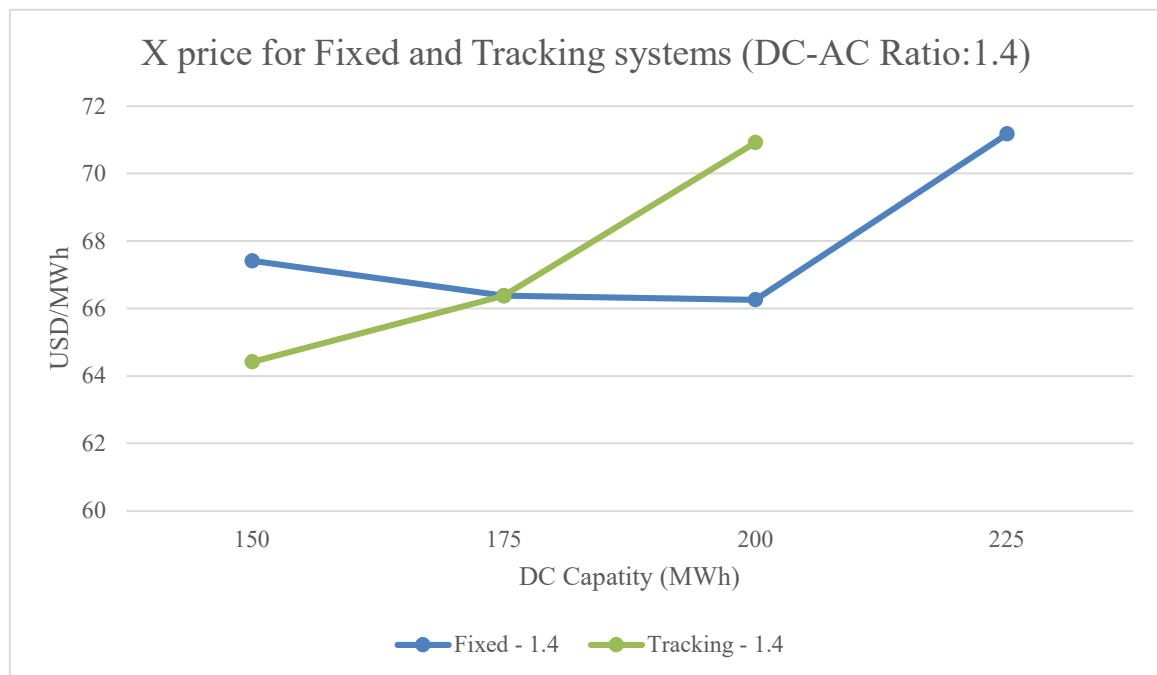
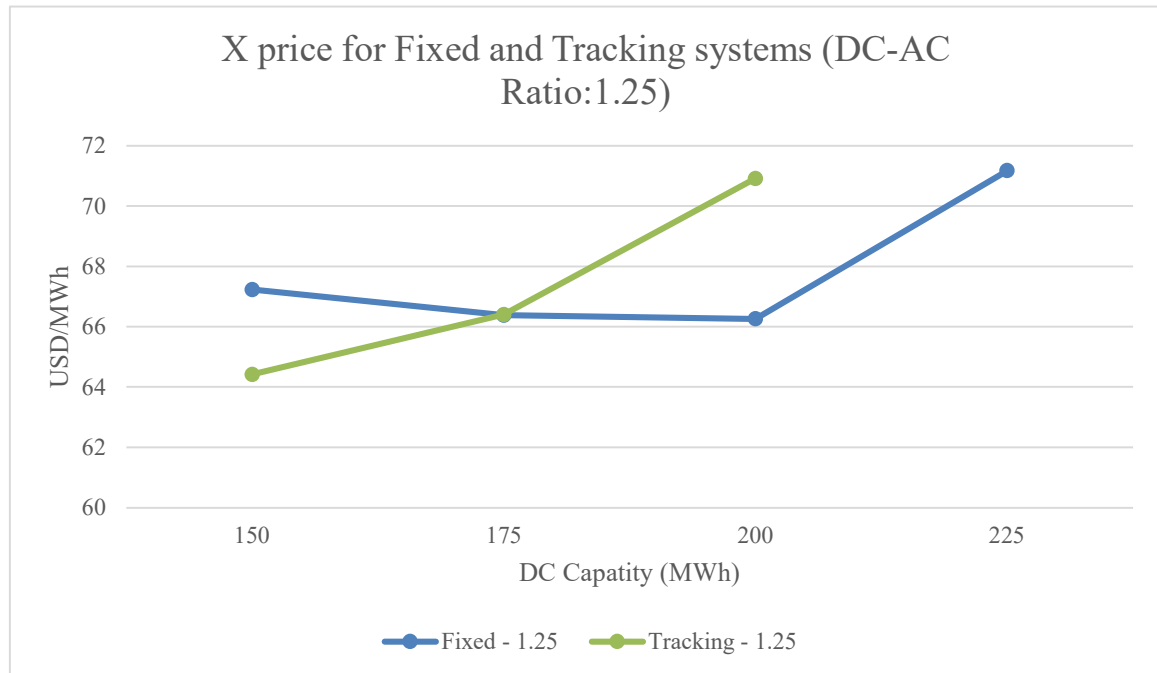
In conclusion, our proposed hybrid solar power plant, consisting of **150 MWdc PV** and **80 MWe CSP**, not only meets all the tender requirements but also delivers a **competitive PPA price of 74.49 USD/MWh**. While the PPA price drove our design strategy, the true challenge lay in securing adequate land for this substantial project while respecting environmental and technical constraints. This initiative offers Brazil a remarkable opportunity to harness its vast solar potential, positioning the country as one of the pioneers in hybrid solar technology. Moreover, it will diversify and strengthen Brazil's predominantly hydroelectric power grid, enhancing energy security and sustainability.

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Appendix A: PV configurations analysis and results

The graphs represent the evolution of the estimated PPA price for the PV plant alone for 2 DC-AC ratio (1.25 and 1.4) and 2 mounting systems (fixed and 1-Axis tracking).



Appendix B: CSP results

Several configurations have been tried for the base design of the CSP part. Here are the full results of the CSP design optimization.

Net Power Output (MW)	Solar Multiple	TES size (h)	X value (USD/MWh)
40	1.2	7	142,92
40	1.4	8	122,86
40	1.6	9	108,40
40	1.8	9	104,23
60	1.2	7	120,71
60	1.4	8	103,95
60	1.6	9	91,88
60	1.8	9	88,74
80	1.2	7	107,41
80	1.4	8	93,25
80	1.6	9	83,38
80	1.8	9	80,34
100	1.2	7	104,44
100	1.4	8	91,49
100	1.6	9	82,91
100	1.8	9	81,60

Appendix C: MATLAB Code used for the dispatch

```
% Constants
max_power_grid = 100; % MWe
evening_peak_hours = 17:23; % Evening peak hours
min_evening_peak_generation_ratio = 0.3; % Minimum generation during evening peak
hours
minimum_power_block = 0.3;
PB_efficiency = 0.38;
g_to_n_efficiency = 0.95;
max_TES = 9; % h
N_days = 365;
N_hours = N_days*24+5;

% Sample hourly generation data for PV and CSP (replace with actual data)
alpha_factor=readmatrix('CCEE - HourlyPriceFactor.xlsx');
PV_generation = alpha_factor(1:N_hours,4); % MW
TOD = alpha_factor(1:N_hours,3);
Thermal_energy = alpha_factor(1:N_hours,5);

% No PV production when the selling price is negative
for h=1:N_hours;
if TOD(h) <=0;
PV_generation(h)=0;
end
End

% Optimization variables
installed_PV_capacity = 150; % MW_AC
installed_CSP_capacity = 80; % MW_el

% Initialize variables to store results
optimal_CSP_generation = zeros(1, N_hours); % To store CSP generation per hour
revenue_per_hour = zeros(1, N_hours); % To store revenue for each hour
total_TES_el = 0 ;
TES_available_per_hour = zeros(1,N_hours);

% Solve optimization problem for each hour
for day=1:N_days;
% Calculation of the total energy harnessed by the heliostat field
% during that day
% NB: The stored energy is calculated as electric generation equivalent
% so its unit is MWh_el
total_TES_el = sum(Thermal_energy(4+(day-
1)*24:day*24+5))*PB_efficiency*g_to_n_efficiency;
if total_TES_el > max_TES*installed_CSP_capacity;
total_TES_el = max_TES*installed_CSP_capacity;
end
TOD_day = TOD((day-1)*24+6:day*24+5);
% Set up for the PB generation of the day
PV_peakhours = sum(PV_generation((day-1)*24+18:day*24));
Total_prod_day = sum(PV_generation((day-1)*24+1:day*24)) + total_TES_el;
prod_peakhours = PV_peakhours;
left_in_peakhours = min_evening_peak_generation_ratio*Total_prod_day -
prod_peakhours;
%Start of the dispatch based on the energy harnessed during the day
while total_TES_el > 0;
% Constraint on each day to reach 30% of total daily prod in peak
% hours : first, only the alpha factors in peak hours are
% considered
if left_in_peakhours > 0;
```

```

% Looking for the best alpha factor in the peak hours range
[max_alpha,max_hour] = max(TOD_day(13:19));
max_hour = max_hour + 13 - 1;
% Defining the maximum prod of the PB for that hour based on the PV prod
max_prod_hour = min(installed_CSP_capacity, 100-PV_generation(24*(day-
1)+max_hour+5));
if total_TES_el >= installed_CSP_capacity;
if (total_TES_el <= (1+minimum_power_block)*installed_CSP_capacity);
% The goal is to send just enough so that the PB can still run on the next hour
optimal_CSP_generation(24*(day-1)+max_hour+5) = min(max_prod_hour, total_TES_el -
minimum_power_block*installed_CSP_capacity);
else
% Maximum power sent to PB when possible
optimal_CSP_generation(24*(day-1)+max_hour+5)= max_prod_hour;
end
elseif total_TES_el >= left_in_peakhours;
% Peak hours usually have lower prices than the middle of
% the night, send just enough to get the 30% of prod, and then switch to other hours
optimal_CSP_generation(24*(day-1)+max_hour+5) = min(max_prod_hour,
left_in_peakhours);
else
% Remaining energy sent if less than max capacity
optimal_CSP_generation(24*(day-1)+max_hour+5)=min(max_prod_hour, total_TES_el);
end
total_TES_el = total_TES_el - optimal_CSP_generation(24*(day-1)+max_hour+5);
TOD_day(max_hour) = 0;
% Update of the share of prod in peak hours for that day
prod_peakhours = prod_peakhours + optimal_CSP_generation(24*(day-1)+max_hour+5);
left_in_peakhours = left_in_peakhours - optimal_CSP_generation(24*(day-
1)+max_hour+5);
else % when 30% of daily prod is reached, all alpha factors can be used
% Looking for the best alpha factor in the whole day
[max_alpha,max_hour] = max(TOD_day);
% Defining the maximum prod of the PB for that hour based on the PV prod
max_prod_hour = min(installed_CSP_capacity, 100-PV_generation(24*(day-
1)+max_hour+5));
if total_TES_el >= installed_CSP_capacity;
if (total_TES_el <= (1+minimum_power_block)*installed_CSP_capacity);
% The goal i to send just enough so that the PB can still run on the next hour
optimal_CSP_generation(24*(day-1)+max_hour+5) = min(max_prod_hour, total_TES_el -
minimum_power_block*installed_CSP_capacity);
else
% Max power sent to PB when possible
optimal_CSP_generation(24*(day-1)+max_hour+5)= max_prod_hour;
end
else
% Remaining energy when not enough
optimal_CSP_generation(24*(day-1)+max_hour+5)=min(max_prod_hour, total_TES_el);
end
total_TES_el = total_TES_el - optimal_CSP_generation(24*(day-1)+max_hour+5);
TOD_day(max_hour) = 0;
end
End

% Printing the amount of TES available at each hour
for hour=4+(day-1)*24:day*24+5;
TES_available_per_hour(hour)=TES_available_per_hour(hour-
1)+Thermal_energy(hour)*PB_efficiency*g_to_n_efficiency-optimal_CSP_generation(hour);
if TES_available_per_hour(hour) > max_TES*installed_CSP_capacity;
TES_available_per_hour(hour) = max_TES*installed_CSP_capacity;
end
end
end

```

```

End

% Calculation of the gross revenues based on the final production
% NB: Revenues are given considering a X price of 1€/MWh and do not take into
% account bonus for firm production (calculated on Excel)
for hour=1:N_hours;
    revenue_per_hour(hour) =
    TOD(hour)*(PV_generation(hour)+optimal_CSP_generation(hour));
end
total_revenues = sum(revenue_per_hour);

% Calculate total PV generation for the day
total_PV_generation = sum(PV_generation);

% Constraint for total CSP generation to be at least 30% of total PV generation
total_CSP_generation = sum(optimal_CSP_generation);
total_gen_peakhours = 0;
for day=1:N_days;
    total_gen_peakhours = total_gen_peakhours + sum(PV_generation((day-1)*24+18:day*24))
    + sum(optimal_CSP_generation((day-1)*24+18:day*24));
end
peakhours_share = total_gen_peakhours/(total_PV_generation+total_CSP_generation);
total_CSP_generation_constraint = total_gen_peakhours >= 0.3 *
(total_PV_generation+total_CSP_generation);
total_prod = optimal_CSP_generation' + PV_generation;

% Display hourly results
fprintf('Optimal CSP generation schedule:\n');
disp('Hour | TOD | CSP_generation | PV_generation | TES available');
disp([(0:N_hours-1)', TOD, optimal_CSP_generation', PV_generation,
TES_available_per_hour']);
fprintf('Total PV generation for the day: %.2f MWh\n', total_PV_generation);
fprintf('Total CSP generation for the day: %.2f MWh\n', total_CSP_generation);
fprintf('Share of generation in peakhours: %.2f \n', peakhours_share*100);

if total_CSP_generation_constraint
    disp('Total peakhours generation meets the constraint (>= 30% of total generation)');
else
    disp('Total peakhours generation does not meet the constraint (>= 30% of total
generation)');
end
fprintf('Total Revenue for the day: %.2f €\n', total_revenues);

% Write the results in an Excel file to use the data for further calculation
writematrix([(0:N_hours-1)', TOD, optimal_CSP_generation', PV_generation, total_prod,
TES_available_per_hour'], 'Results.xls');

```

Appendix D: Implementing linear optimization for hybrid PV-CSP Plant

Linear optimization can be used for energy dispatch. We have a set of energy balances on PV and CSP TES which needs to be balanced every hour. This can be done by defining an objective function (revenue) which needs to be solved for every hour with energy balances as a constraints. Instead of framing 8760 equations for energy balance, the energy balance is solved based on the given constraints and to maximize the objective function (revenue). While solving, the TES storage is updated on available CSP power and excess PV. To verify whether the code is correct, you can see the energy input in the system which should always be equal to output to accumulation (transport equations).

The code is an energy dispatcher capable of operating and controlling the plant as desired according to the KPI or objective function that needs to be optimized based on work of Juan Maz Zapater. The optimization problem can be summarised as follows:

1. Objective function: The mathematical representation of what needs to be maximized or minimized. It is normally made with a set of variables combined to represent the concept that needs to be optimized (profit, production, energy waste...)
2. Constraints: To ensure that the final solution lies within the domain of an acceptable solution, we need to set constraints on our variables. It also acts as a guiding tool to decide to set the variables at some values rather than other ones falling outside the constraints. They are defined using the parameters, which in combination with the variables help create the mathematical expression that will be the constraint.
3. Variables: They are changed around by the optimization tool until it finds the optimal solution to the problem.
4. Parameters: The parameters are fixed values given to the optimization tool to be used for building the constraints. They are not changed by the dispatcher during the total optimization process.

The problem can be defined as follows (Wikander, 2023):

$$\begin{aligned} & \text{maximize or minimize } f(x) = \sum_{i=1}^n c_i x_i \\ & \text{subject to } \sum_{i=1}^n a_{ji} x_i \geq b_j, \forall j = 1 \dots, \end{aligned}$$

Where x_i are the variables and a_{ji} , b_j and c_i are parameters. The first row is the objective function and the second the constraints.

Energy Dispatch Model Layout:

Based on the PV field data, the optimal sizing required for CSP field, the flow diagram looks like below:

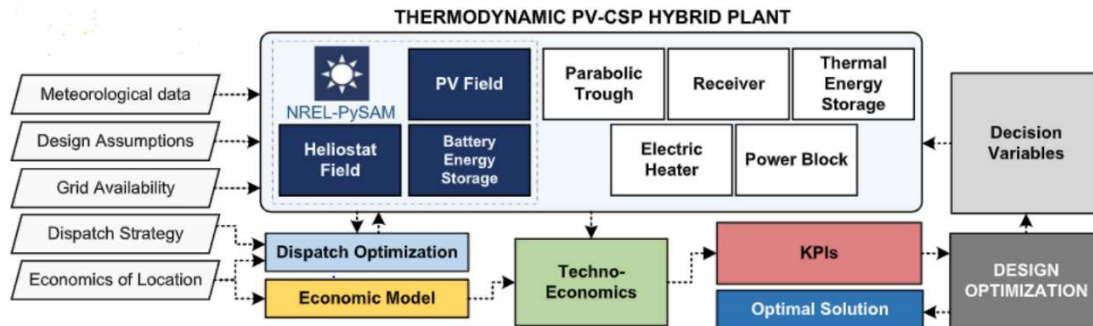


Figure 7: Diagram for Modelling energy dispatch.

The next figure summarizes what the dispatcher will simulate including the variables.

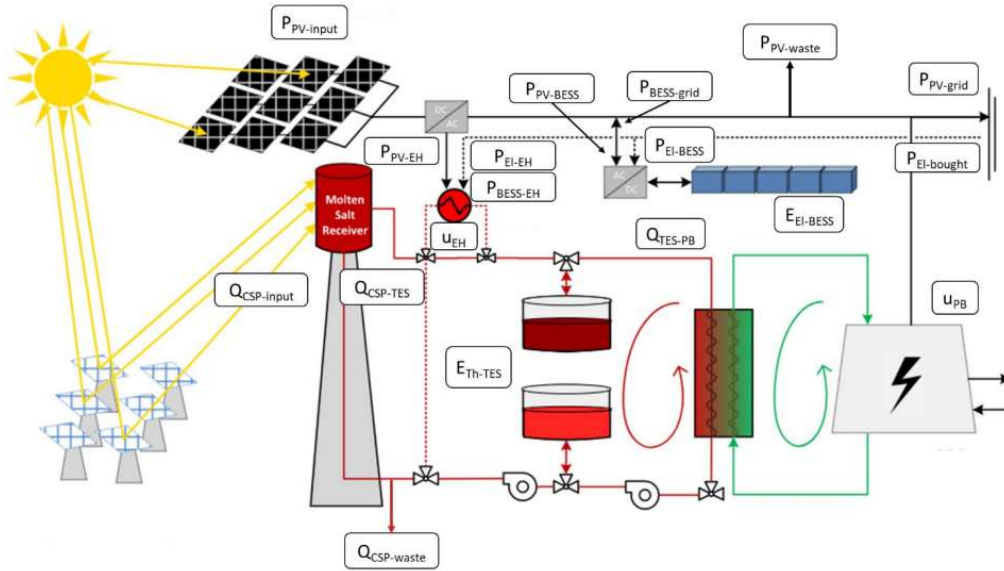


Figure 8: Layout of PV_CSP hybrid power plant.

Tabel 12: Table of variables to be optimized.

Variable	Unit	Definition
PPV-Grid	MW	Power that PV injects to the grid
PPV-EH	MW	Power that PV send to electric heater
PPV-waste	MW	Power from PV that is curtailed
QCSP-TES	MW	Power that CSP system injects to the grid
QTES-PB	MW	Power that CSP system sends to Power Block
PEL-EH	MW	Electricity purchased going to EH
PEH	MW	Total Power going to EH
ETH-TES	MWh _{th}	SoC of the TES
UPB	Binary	Mathematical value representing the state (ON/OFF) of the Power Block

We formulated a set of equations setting constraints for the optimization tool which are the limits of the variables. They are as follows:

The summation is done over every hour step from 1st hour to 8760th hour to simulate energy balance, demand equations over the entire year.

Equation 1: Demand (added as a constraint)

$$Q_{TES_PB}[t] \cdot n_{PB} \cdot n_{HX} + P_{PV_grid}[t] \leq P_{max_injected}$$

The total energy injected from the PV-CSP Hybrid system should not exceed the maximum permissible energy allowed to be injected into the grid.

Equation 2: PV Energy Balance (added as a constraint)

$$P_{PV_grid}[t] + P_{PV_EH}[t] + P_{PV_waste}[t] = P_{PV_input}$$

For every time step, the input energy from PV (i.e. the energy received by the modules) must be equal to the sum of energy injected to the grid, energy sent to the electric heater for storing it as heat and energy curtailed.

Equation 3: CSP System Energy Balance(added as a constraint)

$$Q_{CSP}^{TES}[t] + Q_{waste}^{CSP}[t] = Q_{input}^{CSP}[t]$$

Similarly, for CSP, the energy received by the tower at every time step must be equal to the energy sent to TES and energy curtailed from the CSP.

Equation 4: Electric Heater Energy Balance

$$P_{PV}^{EH}[t] = P_{EH}[t]$$

The electric heater can receive energy from the PV.

Equation 5: Thermal Energy Storage Energy Balance

$$E_{TES}^{TH}[t] = E_{TES}^{TH}[t-1] + P_{EH}[t] + Q_{TES}^{CSP}[t] \cdot n_{rec} - Q_{PB}^{TES}[t]$$

The energy in TES system at every time step is the summation of existing energy and charging/discharging of the TES system through electric heaters, CSP towers and the power block.

Equation 6: State of Charge (SoC) limitation

$$E_{min} \leq E[t] \leq E_{max}$$

The SoC for the TES must always be maintained at a certain level.

Objective functions:

The objective functions are taken from the work of "Analysis of a hybrid PV-CSP plant integration in the Electricity Market by Juan Mazzapater". The objective functions represent the targets the code aims to optimize, whether by minimizing or maximizing them. Since implementation of multiobjective optimization function on MATLAB is difficult, we try to work with a single objective function of maximizing revenue from the PV+CSP Hybrid.

In the context of this project of the hybrid PV-CSP plant, the chosen functions are aligned with the objectives. However, the equations presented here are limited as it doesn't take all the factors utilized in real life hybrid power plant into account.

The following section outlines the objective functions integrated into the energy dispatcher and subjected to testing.

Wasted energy (curtailment) needs to be minimized to run the hybrid system as close to its nameplate capacity as possible, this reduces CAPEX without impacting the revenue.

Objective 1: Maximize the income of the hybrid plant.

$$\bar{F} = \max \sum_1^{8760} (P_{PV_{grid}}[t] + Q_{TES_{PB}}[t] \cdot n_{PB} \cdot n_{HX}) \cdot p_{el_{selling}}[t]$$

Here is the code snippet developed for this project. As explained in Section 6, it is not finished due to a lack of time and the choice to focus on another dispatch strategy targeting only the CSP production.

```
% Constants
max_power_grid = 100; % MW
% initial_CSP_Charge = 0.001; % MWh
max_CSP_Charge = 540; % MWh
min_CSP_Charge = 6; % MWh (Minimum CSP charge remaining)
TOD_per_KWH = 1; % $ per MWh
max_CSP_dispatch = 60;
n_PB = 0.38;
n_gtn = 0.95;
n_el = 0.95;
El_heater_capacity=50; %MWe

N_days = 3; % Enter Number of Days here

N_hours = N_days * 24;

% Sample hourly generation data for PV and CSP (replace with actual data)
alpha_factor = readmatrix('CCEE - HourlyPriceFactor.csv');
PV_generation = alpha_factor(1:N_hours, 4); % MW
TOD_values = alpha_factor(1:N_hours, 2);
Thermal_energy = alpha_factor(1:N_hours, 3);

% Create optimization problem
prob = optimproblem;

% Define variables
CSP_dispatch = optimvar('CSP_dispatch', N_hours, 1, 'LowerBound', 0, 'UpperBound', max_CSP_dispatch);
CSP_charge_remaining = optimvar('CSP_charge_remaining', N_hours, 1, 'LowerBound', 0, 'UpperBound', max_CSP_Charge);
PV_injected = optimvar('PV_injected', N_hours, 1, 'LowerBound', 0, 'UpperBound', max_power_grid);
Thermal_storage = optimvar('Thermal_storage', N_hours, 1, 'LowerBound', 0, 'UpperBound', Thermal_energy);
curatiled_PV = optimvar('curatiled_PV', N_hours, 1, 'LowerBound', 0, 'UpperBound', PV_generation);

% Objective function
objective = -sum(TOD_per_KWH * (CSP_dispatch + PV_injected) .* TOD_values);
prob.Objective = objective;

% Define variable for indicating when to set CSP_dispatch to zero
zero_dispatch = optimvar('zero_dispatch', N_hours, 1, 'Type', 'integer', 'LowerBound', 0, 'UpperBound', 1);

% Add constraint to ensure CSP_dispatch is zero if its value is less than 2
for hour = 1:N_hours
    constraint_name = strcat('CSP_dispatch_nonzero_', num2str(hour));
    prob.Constraints.(constraint_name) = CSP_dispatch(hour) >= 5 * zero_dispatch(hour);
end

% Constraints
constraint1 = (PV_injected + CSP_dispatch) <= max_power_grid;
constraint2 = PV_injected <= PV_generation - curatiled_PV;
constraint3 = CSP_dispatch <= n_PB * n_gtn * CSP_charge_remaining;
constraint4 = CSP_charge_remaining(1) == initial_CSP_Charge(1) - CSP_dispatch(1) + Thermal_storage(1);
constraint5 = CSP_charge_remaining(2:N_hours) == CSP_charge_remaining(1:N_hours - 1) - CSP_dispatch(2:N_hours) + Thermal_storage(2:N_hours) + n_el * curatiled_PV(2:N_hours);
constraint6 = CSP_charge_remaining(2:N_hours) <= max_CSP_Charge;
constraint7 = sum(CSP_dispatch(N_days*24 - 6:N_days*24)) >= 0.3 * (PV_injected(1:N_hours) + CSP_dispatch(1:N_hours));
constraint8 = sum(PV_generation(1:N_hours) + n_PB * n_gtn * Thermal_energy(1:N_hours)) >= sum(CSP_dispatch(1:N_hours) + PV_injected(1:N_hours));
constraint9 = PV_generation == PV_injected + curatiled_PV;
constraint10 = CSP_charge_remaining >= min_CSP_Charge; % New constraint for minimum CSP charge remaining
constraint11 = n_el * curatiled_PV <= El_heater_capacity;

% Add constraints to the problem
prob.Constraints.constraint1 = constraint1;
prob.Constraints.constraint2 = constraint2;
prob.Constraints.constraint3 = constraint3;
prob.Constraints.constraint4 = constraint4;
prob.Constraints.constraint5 = constraint5;
prob.Constraints.constraint6 = constraint6;
prob.Constraints.constraint7 = constraint7;
prob.Constraints.constraint8 = constraint8;
prob.Constraints.constraint9 = constraint9;
prob.Constraints.constraint10 = constraint10;
prob.Constraints.constraint11 = constraint11;

% Solve the optimization problem
options = optimoptions('linprog', 'Display', 'off');
[solution, fval, exitflag] = solve(prob, 'Options', options);
```



```

if exitflag > 0
    optimal_CSP_dispatch = solution.CSP_dispatch;
    optimal_PV_injected = solution.PV_injected;
    CSP_charge_remaining_solution = solution.CSP_charge_remaining;
    Total_power_output = optimal_PV_injected + optimal_CSP_dispatch;
    optimal_curtailment = solution.curtailed_PV;

    % Calculate remaining CSP charge as a percentage of the initial CSP charge
    remaining_CSP_percentage = (CSP_charge_remaining_solution / initial_CSP_Charge) * 100;

    % Calculate revenue for each hour
    revenue_per_hour = TOD_per_KWH * (optimal_PV_injected + optimal_CSP_dispatch) .* TOD_values;

    % Calculate new revenue considering additional constraint
    new_revenue_per_hour = revenue_per_hour;
    for i = 1:N_days
        for j = 3:24
            if solution.CSP_dispatch((i - 1) * 24 + j) > 0 && solution.PV_injected((i - 1) * 24 + j) > 0
                if (solution.CSP_dispatch((i - 1) * 24 + j) + solution.PV_injected((i - 1) * 24 + j)) == ...
                    (solution.CSP_dispatch((i - 1) * 24 + j - 1) + solution.PV_injected((i - 1) * 24 + j - 1)) && (solution.CSP_dispatch((i - 1) * 24 + j - 1) ...
                    + solution.PV_injected((i - 1) * 24 + j - 1)) == (solution.CSP_dispatch((i - 1) * 24 + j - 2) + solution.PV_injected((i - 1) * 24 + j - 2))
                    new_revenue_per_hour((i - 1) * 24 + j) = 1.2 * revenue_per_hour((i - 1) * 24 + j);
                    new_revenue_per_hour((i - 1) * 24 + j - 1) = 1.2 * revenue_per_hour((i - 1) * 24 + j - 1);
                    new_revenue_per_hour((i - 1) * 24 + j - 2) = 1.2 * revenue_per_hour((i - 1) * 24 + j - 2);
                else
                    new_revenue_per_hour((i - 1) * 24 + j) = revenue_per_hour((i - 1) * 24 + j);
                end
            end
        end
    end
end

```

Appendix E: Detailed technical presentation of the final bid

PV		CSP		Hybrid Plant	
PV Module	WAAREE Energies Bifacial model Bi-68-660	Power Block DC capacity	80 MW	Grid capacity	100 MW _{ac}
Inverter	Sichuan Clou Energy Electric Co. NEPCS-2300 [800V]	Solar Multiple	1.8		
DC Capacity	150 MW	TES size	9 hours	Global annual production	459.5 GWh
DC-AC ratio	1.25	HTF	Molten Salt	Capacity Factor	52 %
Mounting system	1-Axis Tracking with 45° rotation	Design Point DNI	1127.93 W/m ²		
Azimuth	North (0°)	Power Block thermal efficiency	0.38		
Tilt angle	-	Heat-Transfer Fluid (HTF) temperature range	Hot: 574°C Cold: 290°C		
Total number of modules	226 920 modules	Solar Field Area	3,796,685 m ²		
Array configuration	31 modules per string, and 7 320 in parallel	Capacity factor	25 %		
Land Area	680,760 m ²				
Capacity factor	27 %				

Appendix F: Financial Analysis of the final bid

This graph represents the cash flow used to calculate the NPV and X price in the final configuration.

